Manufactured Matches: Generative Face Restoration Inflates False Alarms in Open-Set Low-Resolution Recognition

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Abstract. Operators of watchlist face recognition routinely receive low-resolution probes, from distant cameras or compressed video, and a common response is to enhance the probe with a super-resolution or face-restoration model before matching. We measure what this does to open-set identification, where a probe may belong to no enrolled person, using frozen public face encoders and public data. Three results follow. First, no post-hoc processing of a single degraded probe recovers the lost identity: pooling many frames (mean or quality-weighted) saturates at a resolution-set ceiling, and single-image restoration does not cross it. Second, blind restoration actively lowers true identification at every face size from 8 to 48 pixels. Third, and most consequential, restoration guided by a gallery candidate transfers that candidate's identity onto the probe, a vulnerability of the restoration-enabled watchlist pipeline: a non-enrolled stranger is accepted as the candidate 6 to 15 times above the fixed 1% false-alarm baseline, with no gain in true detection. The hazard is worst at the lowest resolutions and resists lightweight mitigation, because a low-resolution probe is itself identity-underdetermined, making a manufactured match indistinguishable from a real one. The single lever that crosses the resolution ceiling is genuine multi-frame super-resolution that fuses real sub-pixel samples, at a measured best-case exchange rate of roughly 8 to 32 frames per resolution step. We give a practitioner rule: do not restore a single low-resolution face before open-set search with a frozen matcher.

Keywords: Open-set face recognition, watchlist identification, low-resolution face recognition, face super-resolution, face restoration, biometric system vulnerability, face morphing and manipulation, generative models, biometric security

1 Introduction

Watchlist face recognition decides, for each incoming probe, whether it belongs to any enrolled identity and, if so, which, while rejecting the far larger stream of non-enrolled people who pass the camera. This open-set decision is harder than verification or closed-set search, and its decisive metric is the true positive identification rate (TPIR) at a fixed false positive identification rate (FPIR). The probes that reach such systems are routinely small: a face spanning a few dozen pixels is the normal case for standoff cameras, compressed video, and crowd surveillance, and resolution at this scale already erodes one-to-many identification [1].

The operational response to a small probe is to repair it before matching. Modern generative face restorers, blind restorers such as GFPGAN [2] and CodeFormer [3], and reference-based diffusion restorers such as ReF-LDM [4], return sharp, confident faces and drop into any pipeline as a one-line preprocessing step. A watchlist naturally invites the reference-based form: the gallery already holds a high-resolution photo of each candidate, so an operator can restore the probe toward its top candidate and re-match.

Super-resolution and restoration have been reported to improve face recognition, but in settings that differ from watchlist deployment: 1:1 verification, closed-set identification, or a recognizer trained jointly with the restorer [5], [6], [7]. Whether enhancement helps or harms open-set identification with a frozen, off-the-shelf matcher, the regime an operator actually runs, has not been measured, and the metric that would expose a problem, the false-identification rate on non-enrolled probes, is not reported by that literature.

We measure it, and find a security vulnerability of the deployed pipeline rather than a mere loss of accuracy. Reference-guided restoration transfers the guiding candidate's identity onto the probe, so a non-enrolled stranger is accepted as that candidate 6 to 15 times above the fixed 1% false-alarm baseline, with no gain in true detection. The failure is invisible to accuracy and rank-1, which score only enrolled probes; it appears only when the operating point is read on strangers. It is also insidious, because the natural source of the guiding reference is the system's own top candidate, so the pipeline manufactures the match with no attacker in the loop. Blind restoration, by contrast, lowers true identification at every face size without inflating false matches, and frame pooling saturates at a resolution-set ceiling that single-image enhancement cannot cross.

Contributions.

(1) We show that reference-guided restoration manufactures matches: restoring a probe toward a gallery candidate transfers that candidate's identity, so strangers are accepted as the candidate 6 to 15 times above baseline at a fixed FPIR, with no detection gain. We show the hazard resists two lightweight defenses and explain why.

(2) We establish a resolution ceiling that explains the hazard: pooling frames saturates below the next resolution step, blind single-image restoration only hurts, and gallery-guided restoration adds no true detection; only genuine multi-frame super-resolution crosses, at a measured frames-for-pixels exchange rate.

(3) We give a practitioner rule and an operating envelope: do not restore a single low-resolution face before open-set search; below about 16 pixels the only safe policy is to abstain.

The paper is organized as follows. Section 2 positions the work against low-resolution recognition, face restoration, and open-set evaluation. Section 3 formalizes the open-set watchlist decision and the restoration pipelines as a threat model. Section 4 details datasets, encoders, degradation, and statistics. Sections 5 to 7 establish the resolution ceiling, the

blind-restoration loss, and the manufactured-match hazard; Section 8 shows the hazard resists lightweight mitigation and Section 9 gives the mechanism. Sections 10 to 12 discuss implications, boundary conditions, and responsible use, and Section 13 concludes.

2 Related work

2.1 Low-resolution and cross-resolution face recognition. Recognition degrades as the face shrinks, and the effect is strongest in one-to-many identification, where a small probe must be ranked against a large gallery [1]. Benchmarks built from genuine surveillance capture this regime directly: TinyFace [8] provides native sub-20-pixel crops, and the QMUL Surveillance Face Recognition Challenge [9] defines the open-set watchlist protocol we adopt. Cross-resolution methods reduce the gap by learning resolution-robust features [10], a training-side remedy that is orthogonal to the frozen-matcher deployment we study.

2.2 Blind face restoration and super-resolution. Generative restorers reconstruct a plausible high-resolution face from a degraded one without a reference. GFPGAN [2] and CodeFormer [3] use a learned facial prior, and Real-ESRGAN [11] targets general blind super-resolution; the field is surveyed in [5]. Multi-frame super-resolution instead fuses several observations of the same face: classical sub-pixel fusion and learned video models such as BasicVSR++ [12] recover real detail, and multi-frame fusion has been shown to raise face identification over single-image super-resolution [13].

2.3 Reference-guided and identity-conditioned restoration. Reference-based restorers condition the reconstruction on one or more identity images. ReF-LDM [4] is a latent-diffusion restorer guided by reference photos; IConFace [14] and FaceMe [15] condition on an identity and report that a mismatched reference makes the model copy the reference's appearance, with reference dominance growing as the input degrades. That line treats reference overuse as an image-fidelity artifact to be reduced; we measure its consequence at the recognition decision.

2.4 Super-resolution for recognition: helps versus hurts. Gains from enhancement appear when the recognizer is trained jointly with or on the restored images, as in cascaded super-resolution with identity priors [6], or when a frozen matcher reads restorations whose identity is preserved under matched synthetic degradation [7]. The opposite is reported in the regime we study, a frozen off-the-shelf matcher on native low resolution: the surveillance challenge finds pre-GAN super-resolution neutral to slightly negative [9], cross-resolution recognition degrades under resolution mismatch [10], forensic analysis finds modern restoration raises the log-likelihood-ratio cost [16], and a recent study finds a learned super-resolution front-end does not beat feeding the aligned low-resolution probe directly [17].

2.5 Open-set and watchlist evaluation. Open-set identification is scored by the true positive identification rate at a fixed false positive identification rate (TPIR at FPIR), the convention of the surveillance challenge [9]. Unlike verification (a 1:1 accept/reject) and closed-set identification (every probe is enrolled), the open-set operating point is set on the non-enrolled stream, which is exactly where the hazard we report becomes visible.

2.6 Identity leakage and hallucination in generative models. Generative upsampling is known to hallucinate identity-changing detail: PULSE [18] shows that a single low-resolution face maps to many sharp, confident, and different reconstructions. Reference conditioning turns this open-endedness into a directed effect, pulling the reconstruction toward a chosen identity [14], [15]. Directed injection of an identity into a frozen matcher has a precedent in face morphing: MIPGAN crafts a morph offline that a verification system accepts as a contributing identity [19]. Our hazard differs on the axes a watchlist cares about: it is produced by an honest restoration pipeline from the probe's own top candidate rather than a crafted morph, it is measured at the open-set operating point rather than in 1:1 verification, and it inflates the false-identification rate on non-enrolled strangers. The manufactured-match hazard is the open-set recognition consequence of that directed pull.

Table 1 contrasts the settings. Prior work that reports a gain uses a trained or non-frozen matcher, a closed-set or verification protocol, or synthetic degradation matched to the restorer; the one open-set surveillance study feeds the low-resolution probe directly. None reports the false- identification rate on non-enrolled probes under modern generative restoration, which is the quantity the manufactured-match hazard lives in.

3 Problem formulation and threat model

Open-set watchlist decision. A gallery enrolls identities, each with one high-resolution photo. For a probe, the system computes a similarity to every gallery entry, takes the top-1 candidate, and accepts the probe as that candidate only if the top score exceeds a threshold; otherwise the probe is rejected as non-enrolled. The threshold is set to a target false positive identification rate (FPIR), the fraction of non-enrolled probes wrongly accepted; the true positive identification rate (TPIR) is the fraction of enrolled probes correctly accepted and identified at that threshold. Accuracy and rank-1 score only enrolled probes and are blind to what happens on the non-enrolled stream, which is where the threshold is actually set.

Restoration as a pre-matching operator. Enhancement inserts an operator on the probe before matching. Blind restoration applies a map that depends only on the probe. Reference-guided restoration applies a map that depends on the probe and a reference image; in a watchlist the natural reference is the probe's own top-1 gallery candidate, because the gallery photo is already in hand. We evaluate both, and also frame pooling and multi-frame super-resolution, which combine several frames of one probe.

Threat model. The hazard requires no attacker. The operator applies a standard, well-intentioned enhancement step, and the guiding reference is the system's own candidate, not an adversarial input. The vulnerability is that this honest pipeline transfers the candidate's identity onto a non-enrolled probe and raises the false-identification rate at the deployed operating point. The same mechanism is available to an adversary who can choose the reference, so the measured

Table 1. The manufactured-match hazard lives in a setting prior work does not measure: a frozen matcher, open-set watchlist, probe-only restoration, scored by the false-identification rate on non-enrolled probes.

setting	frozen matcher	open-set	native LR	probe-only restoration	FPIR on strangers
SR for recognition, gain reported [6], [7]	no / partial	no	no	yes	no
Surveillance / forensic SR studies [9], [16], [17]	yes	partial	yes	yes	no
Reference-based restoration [4], [14], [15]	n/a	no	no	yes	no
Generative identity attack (morphing) [19]	yes	no	no	no	no
This paper	yes	yes	yes	yes	yes

effect is both an operational failure mode and an attack surface for an adversary who can supply the reference.

4 Experimental protocol

Datasets. We build open-set watchlists on three public sets (Table 2). YouTube Faces [20] supplies the synthetic-resolution experiments; ChokePoint [21] supplies real surveillance tracks for multi-frame super-resolution; TinyFace [8] supplies genuine native low resolution. Each enrolled identity contributes a high-resolution gallery photo; known probes come from a different video or session than enrollment; unknown probes come from held-out identities. Per-experiment enrolled and unknown counts are listed in Table 2 rather than carried in the prose.

Encoders. We use two frozen public encoders, ArcFace (InsightFace buffalo_l, R50) [22] and MagFace (R100) [23], the second for a cross-encoder robustness check. Neither is fine-tuned; the deployment we study reads restorations through an off-the-shelf matcher.

Degradation and restorers. Probes are degraded to a controlled ladder (the face resized to 8, 12, 16, 24 or 48 pixels and back to the encoder input) so the only variable is resolution; TinyFace provides the native-resolution confirmation. We evaluate three blind restorers (GFPGAN [2], CodeFormer [3], Real-ESRGAN [11]), one reference-based restorer (ReF-LDM [4]), and one multi-frame super-resolution model (BasicVSR++ [12]), plus mean and quality-weighted frame pooling.

Metrics and statistics. We report rank-1, AUROC between known and unknown probes, and TPIR at fixed FPIR. Cascade thresholds are chosen by two-fold cross-fitting over identities, and confidence intervals are identity-clustered bootstraps. Every comparison within a table or figure is computed in one pass on one split so numbers are directly comparable.

5 The resolution ceiling: frames do not buy pixels

On 1,595 identities (800 enrolled, 795 unknown), mean-pooling the per-frame embeddings of a track raises accuracy with more frames but plateaus by about sixteen frames, far below a single frame one resolution step higher (Table 3; Table 4 and Figure 1 break this out by frame count and resolution).

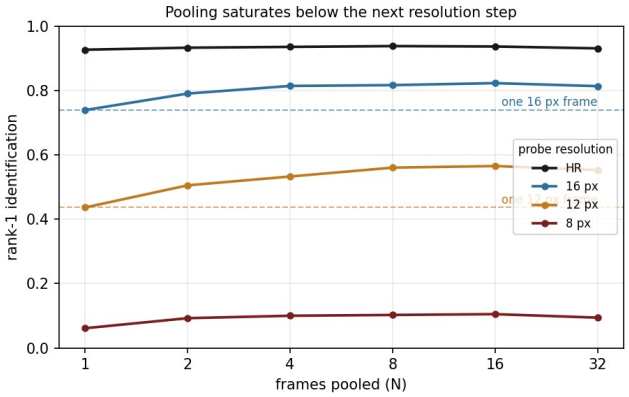


Figure 1. Pooling saturates below the next resolution step. Rank-1 against frames pooled for each probe resolution; dashed lines mark a single frame at the next-higher resolution, which pooling the level below never reaches.

Quality-weighted pooling, weighting frames by the ArcFace magnitude, gives the same answer.

The same saturation holds on ChokePoint surveillance tracks. The one aggregation that does cross the ceiling is genuine multi-frame super-resolution: fusing real sub-pixel-shifted low-resolution samples in pixel space (best case, with known shifts) raises 8 px identification past a single 12 px frame, and 12 px past a single 16 px frame, at an exchange rate of roughly 8 to 32 frames per step (ChokePoint, 15 enrolled and 14 unknown identities, gallery padded with 400 external distractors). Averaging embeddings cannot do this because degradation is a shared bias, not independent variance; only new sub-pixel information crosses.

A modern learned fuser shows the same split on real tracks. Running BasicVSR++[12], a video super-resolution network, on 29 consecutive ChokePoint tracks (15 known, 14 unknown) raises the mated similarity of the restored probe to its true identity monotonically with the number of frames fused (at 8 px, from 0.37 for one frame to 0.50 for sixteen), while the same network given a single frame lowers it below the raw low-resolution frame (0.41 down to 0.37). The dividing line is the source of the added detail: fusing several frames adds real sub-pixel information and helps, consistent with multi-frame face super-resolution raising identification (Ustinova

Table 2. Datasets and per-experiment splits.

dataset	role	enrolled / unknown	resolution
YouTube Faces [20]	pooling ceiling	800 / 795 identities	synthetic 8–48 px
YouTube Faces [20]	blind restoration	400 + 400 ids (800 / 800 probes)	synthetic 8–48 px
YouTube Faces [20]	reference-guided hazard	800 strangers	synthetic 8/12/16 px
ChokePoint [21]	multi-frame SR	15 / 14 (+400 distractors), 29 tracks	real tracks
TinyFace [8]	native-LR blind restoration	native gallery + distractors	native ~20 px

Table 3. Pooling at 8 px never reaches a single 12 px frame (YouTube Faces, ArcFace). Reported at 10% FPIR; at 8 px the 1% point used in Tables 5 and 6 is near the detection floor. Best-N is the best over N up to 32.

metric	best pooling @ 8 px	single frame @ 12 px
rank-1	0.105	0.436
AUROC (known vs unknown)	0.535	0.685
TPIR @ 10% FPIR	0.041	0.282

Table 4. Rank-1 by frames pooled N and probe resolution (YouTube Faces, ArcFace). Each column plateaus with N, below the single-frame value of the column to its left.

N \ resolution	HR	16 px	12 px	8 px
1	0.926	0.739	0.436	0.061
2	0.932	0.790	0.505	0.092
4	0.935	0.814	0.532	0.100
8	0.938	0.816	0.560	0.102
16	0.936	0.823	0.565	0.105
32	0.930	0.813	0.553	0.094

and Lempitsky)[13], whereas single-image restoration invents detail and hurts.

6 Blind restoration lowers identification

We next apply three off-the-shelf restorers (GFPGAN[2], CodeFormer[3], Real-ESRGAN[11]) to the probe and re-match. Across face sizes from 8 to 48 pixels, restoring every probe lowers TPIR at 1% FPIR relative to matching the low-resolution probe directly (Table 5 shows 16 to 48 px; the loss is larger still at 8 and 12 px); the loss shrinks as faces grow but never becomes a gain, and a cross-fitted confidence-gated cascade selects "do not restore." Restoration does not create false alarms here (the false-alarm rate stays near 1%); it destroys true hits by replacing the real face with a sharp, confident, different one.

This holds on genuinely native low resolution, not only synthetic degradation. On TinyFace[8], a surveillance set of native face crops around 20 pixels, matching native probes against a native gallery gives rank-1 0.660 and TPIR at 1% FPIR 0.193; restoring the probes with GFPGAN lowers these to rank-1 0.571 and TPIR 0.013. Restoring the incoming probe moves it off the native gallery distribution, so the watchlist operating point falls sharply.

Table 5. Change in TPIR @ 1% FPIR from restoring every probe, versus the low-resolution baseline (YouTube Faces, 800 enrolled and 800 unknown probes from 400 + 400 identities). Negative is worse. Every cell is a loss.

face size	LR baseline	GFPGAN	CodeFormer	Real-ESRGAN
16 px	0.522	−0.371	−0.266	−0.501
24 px	0.787	−0.240	−0.177	−0.494
48 px	0.868	−0.080	−0.075	−0.045

Table 6. Rate at which a non-enrolled stranger is accepted as its guiding candidate, at a fixed 1% FPIR operating point (YouTube Faces, 800 strangers). Guided restoration is 6 to 15 times the 1% target baseline. The guided cell is the mean with a 95% confidence interval over five independent diffusion-noise seeds.

face size	low-res probe	blind restoration	guided restoration
8 px	0.011	0.001	0.148 [0.139, 0.157]
12 px	0.011	0.001	0.110 [0.104, 0.115]
16 px	0.011	0.001	0.061 [0.056, 0.066]

7 Reference-guided restoration manufactures matches

The sharper hazard appears when restoration is guided by a reference, as a watchlist naturally invites: restore the probe toward its top gallery candidate, then re-match. We use a reference-based latent-diffusion restorer (ReF-LDM[4]) guided by a photo of the candidate from the enrollment video. Guidance transfers the candidate's identity onto the probe (Figure 2). At a threshold calibrated to 1% FPIR on the low-resolution probe, a non-enrolled stranger is accepted as its guiding candidate far above the baseline rate (Table 6).

Guidance buys no true detection to offset this: held at 1% FPIR, guided restoration matches the low-resolution baseline at 8 px and falls below it at 12 and 16 px. A wrong guide pulls the restoration toward the wrong identity about 60% as strongly as the correct guide pulls toward the truth, and equally



Figure 2. The manufactured-match hazard. Each row is one non-enrolled stranger at 8 px: the real low-resolution face, a blind restoration, and the same probe restored toward two different gallery candidates. The restorer returns a sharp, confident face whose identity follows the guide, so a stranger is turned into a match for whichever candidate guides it.

for enrolled and non-enrolled probes, so the restorer confirms whatever candidate it is given.

The hazard is robust in two ways. It persists when the guiding reference is drawn from a separate recording session rather than the enrollment video, at 0.052, 0.078 and 0.020 for 8, 12 and 16 pixel faces, still several times the baseline though milder, so its strength scales with how much the reference resembles the gallery photo. And it transfers across encoders: a second matcher (MagFace[23], a different architecture and loss) reading the same restorations accepts the manufactured identity at matching rates, so the injected identity is not an artifact of one encoder.

8 The hazard resists lightweight mitigation

Two natural gates fail where the hazard is worst. A guidance-boost gate (reject when the guided score exceeds the pre-restoration score by a margin) catches about 90% of manufactured matches but discards 34 to 69% of true hits as resolution drops, because the two populations overlap below 16 px and separate only at 16 px. A guide-promiscuity gate (reject probes that also accept a second random guide) does not separate them at all. The reason is structural: at low resolution a genuine probe is itself compatible with many identities, so a manufactured match is statistically indistinguishable from a real one. The only robust policy below about 16 px is to abstain.

9 Mechanism: the unknownness barrier

All of the above follow from one property of the embedding space, which we call the unknownness barrier: as resolution drops, the embeddings of degraded enrolled probes and of genuine unknowns migrate into the same region, so the known-versus-unknown AUROC falls toward chance. The identity that resolution removes is not recoverable from the pixels of one frame: pooling cannot invent it, single-image restoration can only hallucinate a plausible substitute, and a reference supplies an identity from outside the probe. The exception, genuine multi-frame super-resolution, works precisely because it injects real sub-pixel information that one frame lacked.

10 Discussion and implications

These results reframe pre-match enhancement for watchlist deployment. The dividing line is not whether super-resolution is good or bad, but the source of the detail it adds. Multi-frame super-resolution injects real sub-pixel information from several frames and helps; blind single-image restoration invents detail and lowers true identification; reference-guided restoration imports an identity from the gallery and manufactures matches. Only the first adds information the probe actually contains.

For deployed pipelines the practical consequence is a test, not only a rule. Because the hazard is invisible to accuracy and rank-1, any restoration front-end placed before an open-set matcher should be accepted only after its false-identification rate on non-enrolled probes is measured at the target operating point. We recommend that FPIR on strangers under restoration become a standard acceptance test for watchlist procurement, alongside the usual true-positive reporting, because a front-end that improves rank-1 could at the same time open the manufactured-match channel.

The mechanism also bounds what any post-hoc fix can achieve. Below the resolution where a genuine probe is identity-underdetermined, a manufactured match is statistically indistinguishable from a real one, so no lightweight gate separates them; the robust policy is to abstain and to match the low-resolution probe directly.

11 Limitations

Several boundary conditions scope these results. We report two frozen public encoders, ArcFace (R50) and MagFace (R100, a different architecture and loss); the resolution ceiling of Section 5 and the manufactured-matches hazard of Section 7 both hold for the two. The hazard also holds when the

guiding reference is drawn from a separate recording session rather than the enrollment video, though it is milder, so its strength scales with how much the reference resembles the gallery photo. We use synthetic resolution control across 8 to 48 pixels and confirm the blind-restoration result on native low resolution (TinyFace). The reference-guided hazard is evaluated on the synthetic ladder, not on native sub-20-pixel crops, where the facial landmarks that the 512-input restorer requires are not reliably recoverable; the blind restorers, which detect and align internally, carry the native test. Each reference-guided cell is the mean over five independent diffusion-noise seeds, with 95% confidence intervals in Table 6, over 800 unknown probes that resolve the operating point to about 0.1% FPIR. The pixel-space multi-frame super-resolution result is a best case with known sub-pixel shifts; a learned video super-resolution model (BasicVSR++) on real consecutive tracks confirms the direction below that bound, raising mated similarity monotonically with the number of frames fused.

12 Responsible use and reproducibility

This work is defensive. It documents a failure mode of a common, well-intentioned preprocessing step so that operators can test for it, not a method for attacking a specific deployed system. The reference-guided pipeline we evaluate is standard practice, and the hazard follows from its structure rather than from any product, so we report it in general terms and recommend the acceptance test of Section 10 rather than targeting an implementation.

All experiments use public datasets (YouTube Faces[20], ChokePoint[21], TinyFace[8]) under their published research terms, with public pretrained encoders and restorers; no new face data is collected. Every reported number is reproducible from the project's analysis scripts, which operate on the frozen public encoders and public data above, so the resolution ceiling, the blind-restoration loss, and the manufactured-match hazard can each be regenerated end to end.

13 Conclusion and future work

Enhancing a low-resolution probe before open-set search with a frozen matcher does not recover identity, and when the enhancement is guided by a gallery candidate it manufactures matches, transferring the candidate's identity onto a stranger and inflating false identifications at a fixed operating point. The practitioner rule is to match the low-resolution probe directly and to abstain below about sixteen pixels, where a genuine probe is itself identity-underdetermined. The one enhancement that adds identity rather than inventing or importing it is genuine multi-frame super-resolution, which fuses real sub-pixel detail from several frames. Future work includes measuring the manufactured-match hazard on native sub-20-pixel references, where reliable landmarks for reference-based restorers are not yet available, and standardizing the false-identification-on-strangers acceptance test across restoration front-ends and matchers.

Draft generated from the project's validated experiment findings. Numbers are directly reproducible from the repository's analysis scripts. This HTML is the source of truth; a LaTeX version will be derived from it.

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